

Chapter-6

LAWS OF MOTION AND FRICTION

- If an object is placed in one pan its wt. is ' w_1 ' if placed on another pan its wt. is ' w_2 ' then true wt of object

- If balance has unequal arms, true wt. (w) = $\sqrt{w_1 w_2}$

- If balance has equal arms, true wt. (w) = $\frac{w_1 + w_2}{2}$

- If liquid is flowing through a pipe of cross-section area. A with velocity ' v ' then force needed to hold the pipe:

$$F = v \frac{dm}{dt} = \rho A v^2 \text{ away from himself.}$$

$$\text{If } Q \text{ is discharge, } Q = \frac{\text{volume}}{t} = v A$$

$$F = \frac{\rho Q^2}{A} = \rho A v^2, \quad F \propto v^2 \text{ or } Q^2$$

- Power through pipe, $p = \rho A v^3 = \frac{\rho Q^3}{A^2}$

$$p \propto v^3 \text{ or } Q^3$$

$$\text{If } v_2 = n v_1; p_2 = n^3 p_1$$

- If a force is acting on system and each element of system moves with same acceleration then force on the part of system;

- If there is no force on remaining part after cutting the portion,

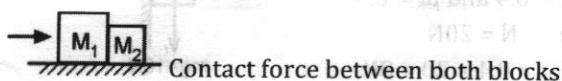
$$F_{\text{part}} = \frac{\text{Remaining mass or length}}{\text{Total mass or length}} \times \text{Force}$$

- If force present on remaining part is in the direction of motion then,

$$F_{\text{part}} = \text{Force} - \frac{\text{Remaining}}{\text{Total}} \times (\text{Force})_{\text{net}}$$

- If force present on remaining part is opposite in the direction of motion then

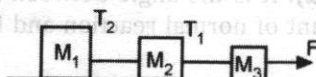
$$F_{\text{part}} = \text{Force} + \frac{\text{Remaining}}{\text{Total}} \times (\text{Force})_{\text{net}}$$



Contact force between both blocks

- If M_1 is cut then contact force = $\frac{M_2}{M_1 + M_2} \times F$

- If M_2 is cut the contact force = $F - \frac{M_1}{M_1 + M_2} \times F$



For T_1 (If m_3 is cut out)

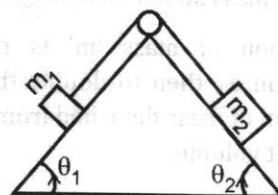
$m_1 + m_2 \Rightarrow$ Remaining mass, $m_1 + m_2 + m_3 = \text{total } m$

And force is not present on remaining part, $T_1 =$

$$\frac{m_1 + m_2}{m_1 + m_2 + m_3} \times F$$

If $m_1 + m_2$ is cut out from system then

$$T_1 = F - \frac{m_3}{m_1 + m_2 + m_3} \times F$$



- Accelⁿ of masses,

$$a = \frac{g}{\text{sum of masses}} (m_2 \sin \theta_2 - m_1 \sin \theta_1)$$

- If $m_2 \sin \theta_2 > m_1 \sin \theta_1$ then m_2 moves down the inclined plane.

- Tension in the strings,

$$T = \frac{\text{product of masses} \times g}{\text{sum of masses}} \times (\sin \theta_1 + \sin \theta_2)$$

- Total force on pulley $F_p = 2T \sin \left(\frac{\theta_1 + \theta_2}{2} \right)$

e.g. (i)

Here, θ_1 and θ_2 both are 90°

($m_2 > m_1$)

$$a = \frac{g}{(m_1 + m_2)} \times (m_2 - m_1)$$

$$T = \frac{m_1 m_2 g}{m_1 + m_2} \times 2$$

$$F_p = 2T$$

(ii) $\theta_1 = 0, \theta_2 = 90^\circ$

$$a = \frac{g}{m_1 + m_2} \times m_2$$

$$T = \frac{m_1 m_2 g}{m_1 + m_2} \times 1$$

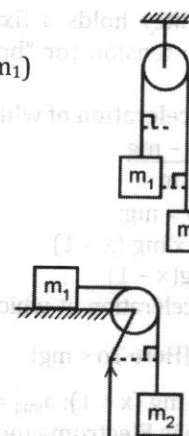
$$F_p = \sqrt{2} T$$

■ Motion of lift

- If lift is accelerating up with acceleration a_0 then

$$\text{App. wt} = \text{Real wt} + m a_0 = m g + m a_0$$

$$\text{App. wt} > \text{Real wt}$$



- (ii) If the lift is accelerating down with acceleration ' a_0 ' then,

$$\text{App. wt} = mg - ma_0 = \text{Real wt} - ma_0$$

$$\text{App. wt} < \text{Real wt.}$$

If cable wire breaks then, $a_0 = g$, App. wt = 0

- (iii) If lift is moving with constant velocity then

$$\text{App. wt} = \text{Real wt (mg)}$$

- If a balloon of mass ' m ' is descending with acceleration a_0 , then for ascending with same acceleration fraction of mass detached from it without changing its volume,

$$\frac{x}{m} = \frac{2a_0}{g + a_0}$$

- If a balloon is ascending with acceleration ' a_0 ' then for descending with same acceleration,

$$\text{fraction of mass added on it, is } \frac{x}{m} = \frac{2a_0}{g - a_0}$$

- If a balloon of mass ' m ' is rising up with acceleration a_0 , then to double the acceleration, the fraction of mass detached from it is without it changing its volume.

$$\frac{x}{m} = \frac{a_0}{g + 2a_0}$$

- If a man standing on platform of weighing machine takes quick step, then his weight first decreases and then increases.
- If a stone of mass ' m ' dropped from height h penetrates distance ' s ' into sand then.

- (i) Average retardation offered by sand
- $$= g \frac{h}{s}$$

- (ii) time of penetration $t = \sqrt{\frac{2s^2}{gh}}$

- (iii) Average resistance offered by sand
- $$= mg \left(1 + \frac{h}{s} \right)$$

- If a monkey holds a fixed vertical wire, having tolerable tension (or "breaking strength") is ' T_0 ' then

- (i) max^m acceleration of which monkey climb up

$$a_{\max} = \frac{T_0 - mg}{m}$$

Here, $T_0 > mg$;

$$\text{if } T_0 = x mg \ (x > 1)$$

$$a_{\max} = g(x - 1)$$

- (ii) Min^m acceleration at which it can slide down, $a_{\min} =$

$$\frac{mg - T_0}{m} \quad [\text{Here } T_0 < mg]$$

$$\text{if } T_0 = x mg \ (x < 1): a_{\min} = g(1 - x)$$

- Friction force is Electromagnetic force.
- Static friction is also known as self adjust opposing force.

Value of static friction, $f_{\text{static}} = f_{\text{applied}}$

- Limiting friction

Condⁿ: Rest into limiting condition or just ready to slide or just ready to move

- Limiting friction force is maximum friction force

- $f_{\text{lim}} \propto \text{Normal Reaction (N)}$

$f_{\text{lim}} = \mu_s N$, where μ_s is constant known coefficient of static friction, which depends upon nature of contact surface does not depend upon normal reaction.

- Kinetic friction or Dynamic friction

Condⁿ: when motion occurs.

$$f_k \propto \text{Normal reaction (N)}$$

$f_k = \mu_k N$; μ_k = coefficient of kinetic friction which depends upon nature of contact surface. It does not depends upon normal reaction and degree of relative motion.

- As, $f_{\text{lim}} > f_k$; $\mu_s > \mu_k$

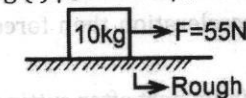
- For analysis of friction force on the body we calculate limiting friction as, $f_{\text{lim}} = \mu_s N$ and compare with applied force.

- (i) If Applied force $< f_{\text{lim}}$, then static friction acts on the body $f_{\text{static}} = f_{\text{applied}}$

- (ii) If Applied force $= f_{\text{lim}}$, then body just move i.e., limiting friction acts on the body, $f_{\text{lim}} = \mu_s N$.

- (iii) If Applied force $> f_{\text{lim}}$, then motion occurs i.e., kinetic friction acts on the body $f_k = \mu_k N$

e.g (i) $\mu_s = 0.6$ & $\mu_k = 0.5$



Friction force on the block = ?

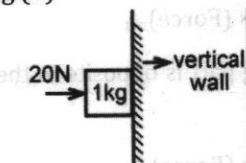
Solⁿ:

$$N = mg = 100N$$

$$f_{\text{lim}} = 0.6 \times 100 = 60$$

$$f < f_{\text{lim}}, f_{\text{static}} = 55N$$

eg.(ii)



$$\mu_s = 0.4 \text{ and } \mu_k = 0.35$$

Solⁿ: $N = 20N$

$$f_{\text{lim}} = 0.4 \times 20 = 8N$$

In this case 'mg' appear as applied force,

Applied force = 10N, App. force $> f_{\text{lim}}$

$$f_k = \mu_k N = 0.35 \times 20 = 7N$$

Angle of friction (α): It is the angle between reaction and resultant of normal reaction and friction force.

(i) If body is at Rest then, $\tan \alpha = \frac{f_{\text{static}}}{N}$

(ii) If body is at **Rest into limiting condition** or just **ready to slide** then, $\tan \alpha = \frac{f_{\text{lim}}}{N} = \mu_s$

(iii) If body is in motion then $\tan \alpha = \frac{f_k}{N} = \mu_k$

▪ Maximum friction angle, $\alpha_{\text{max}} = \tan^{-1}(\mu_s)$

♦ Angle of Repose (θ_R) is the minimum angle of inclination of inclined plane with horizontal such that a body placed on it, just begins to slide.

$$\tan \theta_R = \mu_s$$

$$\therefore \theta_R = \alpha_{\text{max}}$$

▪ Acceleration of a body in a rough inclined plane of inclination ' θ '.

(i) for downward motion, $a = g(\sin \theta - \mu_k \cos \theta)$

(ii) for upward motion, $a = -g(\sin \theta + \mu_k \cos \theta)$

▪ If a body is allowed to slide down from top smooth inclined plane of inclination ' θ ' then body takes ' t ' time to reach the bottom, now the inclined plane is roughed then body takes ' nt ' ($n > 1$) to reach the bottom then the coefficient of friction.

$$\mu_k = \tan \theta \left(1 - \frac{1}{n^2} \right)$$

If t_s = time for smooth inclined plane

t_r = time for rough inclined plane

$$\mu_k = \tan \theta \left(1 - \frac{t_s^2}{t_r^2} \right)$$

▪ When a uniform chain (or uniform rope) of length ' L ' is overhung from the edge of rough horizontal table, having friction coefficient ' μ ' then (i) maximum of its hanging length;

$$x = \frac{\mu L}{\mu + 1}$$

(i) Max^m % of its length = $\frac{100 \mu}{\mu + 1}$

(ii) If $\frac{1}{n}$ of its length should hanged then the coefficient of friction,

$$\mu = \frac{1}{n - 1}$$

▪ If a block of mass ' m ' is placed on rough inclined pane of inclination ' θ ', its downward motion can be prevented slide by applying an upward pull ' F ' and it can be made to slide upwards by a force ' nF ' then the coefficient of friction:

$$\mu = \left(\frac{n - 1}{n + 1} \right) \tan \theta$$

▪ If a block is moving with velocity ' u ' placed on rough horizontal surface having coefficient of

friction ' μ_k ' then distance moved by block before coming to rest,

$$s = \frac{u^2}{2 \mu_k g}$$

▪ Pulling is easier than pushing because pulling decrease the normal reaction.

▪ To avoid slipping while walking on ice, one should take smaller steps because of small a friction.

▪ Direction of friction force on bicycle wheel

(i) when bicycle is peddled friction force on rear wheel is **forward** direction and on front wheel in **backward**.

(ii) When bicycle is not peddled then friction force on both wheel is in **backward** direction.

▪ If a block of mass ' m ' is placed on rough horizontal surface, the coefficient of friction between block and surface is ' μ ' then least force required to slide the block is:

$$F_{\text{min}} = \frac{\mu mg}{\sqrt{1 + \mu^2}}$$

▪ If friction angle is ' α ' then least force required is

$$F_{\text{min}} = mg \sin \alpha$$

▪ Min^m horizontal force or min^m force along the plane required to move the block.

$$F_{\text{min}} = \mu mg = mg \tan \alpha$$

▪ If the coefficient of friction between an insect and hemispherical bowl having radius ' r ' is ' μ ' then insect can crawl upto max^m height:

$$h = r \left[1 - \frac{1}{\sqrt{1 + \mu^2}} \right]$$

Multiple Choice Questions

1. A balloon of gross weight w newton is falling vertically downward with a constant acceleration $a(<g)$. The magnitude of the air resistance is:

- a. w b. $w \left(1 + \frac{a}{g} \right)$
c. $w \left(1 - \frac{a}{g} \right)$ d. $w \frac{a}{g}$

2. A block of mass m is placed on a smooth plane inclined at an angle θ with the horizontal. The force exerted by the plane on the block has a magnitude:

- a. mg b. $mg \sec \theta$
c. $mg \cos \theta$ d. $mg \tan \theta$

3. A block of weight 5 N is pushed against a vertical wall by a force 12 N. The coefficient of friction between the wall and block is 0.6. The magnitude of the force exerted by the wall on the block is:

- a. 12 N b. 5 N
c. 7.2 N d. 13 N

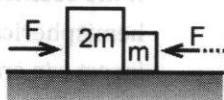
4. A block is gently placed on a conveyor belt moving horizontally with constant speed. After $t = 4$ s, the velocity of the block becomes equal to the velocity of the belt. If the coefficient of friction between the block and the belt is $\mu = 0.2$ then the velocity of the conveyor belt is:

- a. 2 ms^{-1} b. 4 ms^{-1}
c. 64 ms^{-1} d. 8 ms^{-1}

5. An elevator and its load have a total mass of 800 kg. If the elevator, originally moving downward at 10 m/s, is brought to rest with constant deceleration in a distance of 25 m, the tension in the supporting cable will be: ($g = 10 \text{ m/s}^2$)

- a. 94 N b. 1880 N
c. 940 N d. 9600 N

6. Two blocks are in contact on a frictionless table. One has mass m and the other $2m$. A force F is applied on $2m$ as shown in the figure. Now the same force F is applied from the right on m . In the two cases respectively, the force of contact between the two blocks will be



- a. same b. 1 : 2
c. 2 : 1 d. 1 : 3

7. A body is projected up a 45° rough incline. If the coefficient of friction is 0.5, then the retardation of the block is:

- a. $\frac{g}{2\sqrt{2}}$ b. $\frac{g}{\sqrt{2}}$
c. $\frac{3g}{2\sqrt{2}}$ d. $\frac{g}{2}$

8. A fireman wants to slide down a rope. The breaking load for the rope is $\frac{3}{4}$ th of the weight of the man.

With what minimum acceleration should the fireman slide down:

- a. $\frac{g}{3}$ b. $\frac{g}{6}$
c. $\frac{g}{4}$ d. $\frac{g}{2}$

9. A body of mass 10 kg is placed on rough surface, pulled by a force F making an angle of 30° to the horizontal. If the angle of friction is also 30° then the minimum magnitude of force F required to move the body is equal to: (take $g = 10 \text{ m/s}^2$)

- a. 100 N b. $50\sqrt{2}$ N
c. $100\sqrt{2}$ N d. 50 N

10. The coefficient of friction between the tyres and road is 0.4. The minimum distance covered before attaining a speed of 8 ms^{-1} starting from rest is nearly: ($g = 10 \text{ m/s}^2$)

- a. 8.0 m b. 4.1 m
c. 16.4 m d. 18.3 m

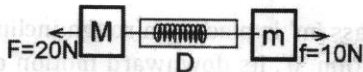
11. The breaking strength of the cable used to pull a body is 40 N. A body of mass 8 kg is resting on a table of $\mu = 0.20$. The maximum acceleration which can be produced by the cable is (Take $g = 10 \text{ ms}^{-2}$):

- a. 6 ms^{-2} b. 3 ms^{-2}
c. 8 ms^{-2} d. 5 ms^{-2}

12. A mass of 3 kg descending vertically downward support a mass of 2 kg by means of a light string passing over a pulley. At the end of 5 second the string breaks. How much higher the 2 kg mass will go?

- a. 4.9 m b. 9.8 m
c. 19.6 m (d) 2.45 m

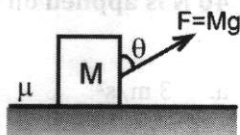
13. A dynamometer D is attached to two bodies of masses $M = 6 \text{ kg}$ and $m = 4 \text{ kg}$. Forces $F = 20 \text{ N}$ and $f = 10 \text{ N}$ are applied to the masses as shown. The dynamometer reads:



- a. 10 N b. 20 N
c. 6 N d. 14 N

14. A block of mass M rests on a rough horizontal surface as shown. Coefficient of friction between the block and the surface is μ . A force $F = Mg$ acting

at angle θ with the vertical side of the block pulls it. In which of the following cases the block can be pulled along the surface:



- a. $\tan\theta \geq \mu$ (b) $\tan\left(\frac{\theta}{2}\right) \geq \mu$
c. $\cot\theta \geq \mu$ d. $\cot\left(\frac{\theta}{2}\right) \geq \mu$

15. A 13m ladder is placed against a smooth vertical wall with its lower end 5 m from the wall. What should be the coefficient of friction between ladder and floor so that it remains in equilibrium.

- a. 0.1 b. 0.15
c. 0.21 d. 0.22

16. A body is released from the top of a smooth inclined plane of inclination θ . It reaches the bottom with velocity v . If the angle of inclination is doubled, what will be the velocity of the body on reaching the ground:

- a. v b. $2v$
c. $[2\cos\theta]^{1/2}v$ d. $[2\sin\theta]^{1/2}v$

17. A box of mass 8 kg is placed on a rough inclined plane of inclination θ . Its downward motion can be prevented by applying an upward pull F . And it can be made to slide upwards by applying a force $2F$. The coefficient of friction between the box and the inclined plane is:

- a. $\frac{1}{3}\tan\theta$ b. $3\tan\theta$
c. $\frac{1}{2}\tan\theta$ d. $2\tan\theta$

18. A block is placed on the top of a smooth inclined plane of inclination θ kept on the floor of a lift. When the lift is descending with a retardation a , the block is released. The acceleration of the block relative to the incline is:

- a. $g\sin\theta$ b. $a\sin\theta$
c. $(g-a)\sin\theta$ d. $(g+a)\sin\theta$

19. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half rough. A block starting from rest at the top of the plane will

again come to rest at the bottom if the coefficient of friction between the block and the lower half of the plane is given by:

- a. $\mu = 2\tan\theta$ b. $\mu = \tan\theta$
c. $\mu = \frac{2}{\tan\theta}$ d. $\mu = \frac{1}{\tan\theta}$

20. A body is sliding down a rough inclined plane of angle of inclination θ for which the coefficient of friction varies with distance x as $\mu(x) = kx$, where k is a constant. Here x is the distance moved by the body down the plane. The net force on the body will be zero at a distance x_0 given by:

- a. $\frac{\tan\theta}{k}$ b. $k\tan\theta$
c. $\frac{\cot\theta}{k}$ d. $k\cot\theta$

21. An object is resting at the bottom of two strings which are inclined at an angle of 120° with each other. Each string can withstand a tension of 20N. The maximum weight of the object that can be sustained without breaking the string is:

- a. 10 N b. 20 N
c. $20\sqrt{2}$ N d. 40 N

22. A rough vertical board has an acceleration a along the horizontal so that a block of mass M pressing against it does not fall. The coefficient of friction between block and the board is:

- a. $> \frac{a}{g}$ (b) $< \frac{a}{g}$
c. $= \frac{a}{g}$ d. $> \frac{g}{a}$

23. A smooth inclined plane of length L having inclination θ with the horizontal is inside a lift which is moving down with a retardation a . The time taken by a body to slide down the inclined plane from rest will be:

- a. $\sqrt{\frac{2L}{(g+a)\sin\theta}}$ b. $\sqrt{\frac{2L}{(g-a)\sin\theta}}$
c. $\sqrt{\frac{2L}{a\sin\theta}}$ d. $\sqrt{\frac{2L}{g\sin\theta}}$

24. The velocity of a bullet is reduced from 200 m/s to 100 m/s while travelling through a wooden block

of thickness 10 cm. Assuming it to be uniform, the retardation will be:

- a. $15 \times 10^5 \text{ m/s}^2$ b. $10 \times 10^4 \text{ m/s}^2$
c. $12 \times 10^4 \text{ m/s}^2$ d. 14.5 m/s^2

25. A bird is sitting in a large closed cage which is placed on a spring balance, it records a weight of 35 N. The bird (mass = 0.5 kg) flies upward in the cage with an acceleration of 2 m/s^2 . The spring balance will now record a weight of:

- a. 27 N b. 35 N
c. 26 N d. 24 N

26. An open knife edge of mass m is dropped from a height 'h' on a wooded floor. If the blade penetrates upto the depth d into the wood, the average resistance offered by the wood to the knife edge is:

- a. $mg \left(1 + \frac{h}{d}\right)$ b. $mg \left(1 + \frac{h}{d}\right)^2$
c. $mg \left(1 - \frac{h}{d}\right)$ d. mg

27. A car is moving along straight horizontal road with a speed v_0 . If the coefficient of friction between the tyres and the road is μ . The shortest distance in which the car can be stopped is: [IOE]

- a. $\frac{v_0^2}{\mu}$ b. $\left(\frac{v_0}{\mu g}\right)^2$
c. $\frac{v_0^2}{\mu g}$ d. $\frac{v_0^2}{2\mu g}$

28. Pushing force making an angle θ to the horizontal is applied on a block of weight W placed on a horizontal table. If the angle of friction be ϕ , the magnitude of force required to move the body is equal to:

- a. $\frac{W \cos \phi}{\cos(\theta + \phi)}$ b. $\frac{W \sin \phi}{\cos(\theta + \phi)}$
c. $\frac{W \tan \phi}{\sin(\theta + \phi)}$ d. $\frac{W \sin \phi}{\tan(\theta + \phi)}$

29. A block B of mass 5 kg is placed on another block A of mass 10 kg, which rests on a smooth horizontal surface. If $\mu = 0.4$ between A and B and a force $F =$

40 N is applied on block B, the acceleration of A is:

- a. 3 m/s^2 b. 2 m/s^2
c. 4 m/s^2 d. $\frac{8}{3} \text{ m/s}^2$

30. A block of mass 4 kg is placed on a rough horizontal plane. A time dependent horizontal force $F = kt$ acts on the block, $k = 2 \text{ N/s}^2$. The frictional force between the block and plane at time $t = 2 \text{ sec}$ is:

- ($\mu = 0.2$)
a. 4 N b. 8 N
c. 12 N d. zero

Answers:

1. c	2. c	3. d	4. d	5. d
6. b	7. a	8. c	9. d	10. a
11. b	12. a	13. d	14. d	15. c
16. c	17. a	18. d	19. a	20. a
21. b	22. d	23. a	24. a	25. b
26. a	27. d	28. b	29. b	30. a

Solution for Selected Problems

1. $mg - R_a = ma$

$$R_a = mg - ma = w - \frac{w}{g} a = w \left(1 - \frac{a}{g}\right)$$

3. $f_{\text{lim}} = 0.6 \times 12 = 7.2 \text{ N}$

weight = 5N

$f_{\text{lim}} > \text{weight}$

Hence friction force = 5N

force exerted by wall on block = $\sqrt{12^2 + 5^2}$

= 13N

4. acceleration of block = $mg = 0.2 \times 10 = 2$

velocity after 4 sec, $v = u + at = 0 + 2 \times 4$

$v = 8 \text{ m/sec}$

5. Retardation of elevator, $\frac{10^2}{2 \times 25} = \frac{100}{50} = 2 \text{ m/s}^2$

Tension in cable, $T = mg + m \times (-a)$

= $800 \times 10 + 800 \times 2$

= 9600N

6. When force F is applied on mass $2m$ then contact force,

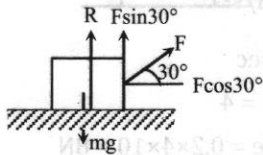
$$P_1 = \frac{m}{m+2m} \times F = \frac{F}{3}$$

when force F is applied on mass ' m ' then contact force

$$P_2 = \frac{2m}{m+2m} \times f = \frac{2f}{3}$$

Ratio of P_1 and $P_2 = 1 : 2$

9.



$$\mu = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$R = mg - F \sin 30^\circ$$

$$F \cos 30^\circ \geq \mu R$$

$$F_{\min} = \frac{\mu mg}{\cos 30^\circ + \mu \sin 30^\circ}$$

$$= \frac{\frac{1}{\sqrt{3}} \times 10 \times 10}{\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} \times \frac{1}{2}} = \frac{2 \times 10 \times 10}{4} = 50 \text{ N}$$

$$11. a = \frac{(T - f_k)}{m} = \frac{(40 - 0.2 \times 8 \times 10)}{8} = 3 \text{ m/s}^2$$

$$12. a = \left(\frac{3-2}{3+2} \right) \times 9.8 = 1.96 \text{ m/s}^2$$

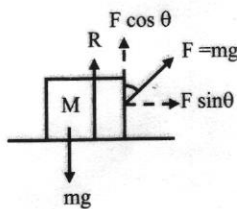
velocity after 5 sec; $V = 1.96 \times 5 = 9.8 \text{ m/sec}$

$$\text{height } h = \frac{V^2}{2g} = \frac{9.8^2}{2 \times 9.8} = 4.9 \text{ m}$$

13. Reading on dynamometer

$$R = 20 - \left(\frac{6}{6+4} \right) \times 10 = 20 - 6 = 14 \text{ N}$$

14.



$$R = mg - mg \cos \theta = mg(1 - \cos \theta)$$

$$= 2 \sin^2 \frac{\theta}{2} \times mg$$

$$f_{\lim} = \mu R = 2 \mu mg \sin^2 \frac{\theta}{2}$$

$$\therefore F \sin \theta \geq F_{\lim}$$

$$\Rightarrow mg \sin \theta \geq 2 \mu mg \sin^2 \frac{\theta}{2}$$

$$\therefore \cot \frac{\theta}{2} \geq \mu$$

15.

$$\cos \theta = \frac{5}{13}$$

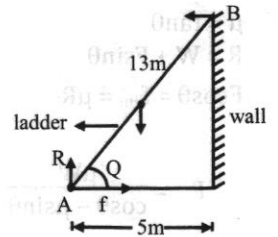
$$f = \mu R \text{ and } R = W$$

$$\Sigma \tau_B = 0$$

$$\Rightarrow R \times 5 - W \times 2.5 - \mu R \times 12 = 0$$

$$12 \times \mu R = W \times 25 \Rightarrow \mu =$$

$$\frac{2.5}{12} = 0.21$$



19. Acceleration of smooth inclined plane

$$a = g \sin \theta$$

Acclⁿ on rough inclined plan,

$$a = g(\sin \theta - \mu_k \cos \theta)$$

$$\text{velocity after half length, } V^2 = 2 g \sin \theta \times \frac{\ell}{2}$$

Again, at bottom, block comes into rest then,

$$0 = V^2 + 2 g(\sin \theta - \mu_k \cos \theta) \times \frac{\ell}{2}$$

$$\Rightarrow 0 = 2 g \sin \theta \times \frac{\ell}{2} + 2 g(\sin \theta - \mu_k \cos \theta) \times \frac{\ell}{2}$$

$$\therefore \mu_k = 2 \tan \theta$$

20. Net force on the block which is sliding down;

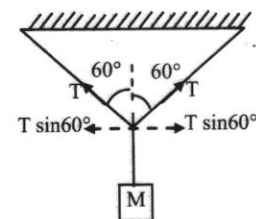
$$F_{\text{net}} = mg \sin \theta - \mu mg \cos \theta$$

$$F_{\text{net}} = 0$$

$$\mu = \tan \theta$$

$$\Rightarrow kx = \tan \theta \therefore x = \frac{\tan \theta}{k}$$

21.

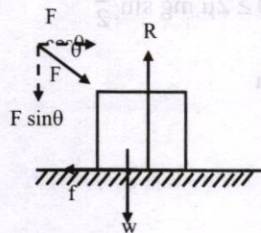


$$2T \cos 60^\circ = w$$

$$\therefore w = 20 \text{ N}$$

$$24. \text{Retardation, } (-a) = \frac{200^2 - 100^2}{2 \times 10 \times 10^{-2}} = 15 \times 10^5 \text{ m/s}^2$$

28.



$$\mu = \tan \theta$$

$$R = W + F \sin \theta$$

$$F \cos \theta = f_{\text{lim}} = \mu R$$

$$F = \frac{\mu W}{\cos \theta - \mu \sin \theta} = \frac{\frac{\sin \theta}{\cos \theta} \cdot W}{\cos \theta - \frac{\sin \theta}{\cos \theta} \sin \theta}$$

$$\therefore F = \frac{W \sin \phi}{\cos(\theta + \phi)}$$

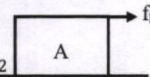
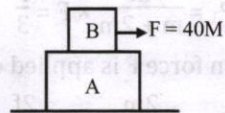
$$29. m_B = 5 \text{ kg}$$

$$m_A = 10 \text{ kg}$$

$$f_k = 0.4 \times 5 \times 10 = 20 \text{ N}$$

accⁿ of block A,

$$a_A = \frac{f_k}{m_A} = \frac{20}{10} = 2 \text{ m/s}^2$$



30. Applied force after t = 2 sec

$$F = 2 \times 2 = 4$$

$$(f_{\text{lim}}) \text{ max}^m \text{ opposing force} = 0.2 \times 4 \times 10 = 8 \text{ N}$$

$$f_{\text{lim}} > F$$

$$\therefore \text{friction force} = 4 \text{ N}$$

